Vivekanand Education Society's Institute of Technology



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Capstone Project – I

Synopsis (2025–26) – Sem VI

Title : Genotype Inference from Leaf Phenotype Using Venation Patterns and Deep Learning

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Abstract

Leaves are one of the most accessible plant organs and exhibit distinct morphological characteristics such as shape, texture, and venation patterns. Leaf venation represents the vascular structure of a leaf and is a stable phenotypic trait that is strongly influenced by genetic regulation. Due to its consistency across environmental conditions, venation serves as a reliable feature for plant characterization.

Direct genotypic analysis of plants requires molecular techniques that are expensive and impractical for large-scale use. Phenotype-based analysis provides a non-invasive alternative by utilizing observable traits to infer underlying genetic information. This project proposes a deep learning-based approach to infer genotype-associated information from leaf phenotypes using venation patterns of Indian medicinal plants. Convolutional Neural Networks (CNNs) are used to automatically learn venation-driven phenotypic features from leaf images and map them to genotype-related classes such as species or genetic groups. The proposed system aims to support automated plant phenotyping and medicinal plant authentication.

Introduction

The genotype–phenotype relationship is a central concept in biology and genetics. While genotype determines the genetic blueprint of an organism, phenotype represents the physical expression of this blueprint influenced by environmental conditions. In plants, many phenotypic traits such as leaf shape, size, texture, and venation are under strong genetic control. Among these, leaf venation is a particularly stable and informative trait, reflecting conserved genetic pathways involved in vascular development.

India is one of the world's largest reservoirs of medicinal plant diversity. Correct genetic identification of these plants is essential for medicinal authenticity, conservation, and breeding programs. However, conventional genotyping techniques such as DNA sequencing, PCR, and molecular markers are costly, time-consuming, and require laboratory infrastructure. This limits their applicability in large-scale or field-based scenarios.

Phenotype-based inference provides a practical alternative by utilizing observable traits as proxies for genetic information. With advances in computer vision and deep learning, it is now possible to automatically extract complex phenotypic patterns from images. Convolutional Neural Networks (CNNs) have demonstrated remarkable success in learning hierarchical representations that correlate strongly with genetic variation. This project explores the feasibility of inferring genotype-related information from leaf venation phenotypes using deep learning techniques.

Problem Statement

Direct genotypic analysis of medicinal plants requires molecular techniques that are expensive, invasive, and impractical for large-scale deployment. Existing image-based plant identification systems focus mainly on species recognition and do not explicitly address genotype–phenotype associations. Furthermore, many models rely on general visual

features and underutilize venation patterns, which are genetically regulated. There is a need for an automated system that can infer genotype-associated information from leaf phenotypes, particularly venation patterns, using deep learning.

Objectives of the Project

- To extract venation-driven phenotypic features from leaf images.
- To study the relationship between leaf venation phenotype and plant genotype.
- To design a CNN-based model for genotype inference from phenotypic traits.
- To map learned phenotypic representations to genotype-associated labels such as species or genetic groups.
- To evaluate the effectiveness of phenotype-based genotype inference.

Proposed Solution

The proposed solution is an automated deep learning-based system that infers genotype-associated information from leaf phenotypes, with a primary focus on venation patterns. The system utilizes image preprocessing techniques to enhance venation structures and suppress background noise. A convolutional neural network (CNN) is employed to automatically learn venation-driven phenotypic features such as vein density, branching patterns, and vascular topology from leaf images. These learned features are then mapped to genotype-associated classes, such as species or genetic groups. The proposed approach provides a non-invasive, scalable, and efficient alternative to conventional molecular genotyping methods.

Methodology

1. Dataset Collection

The dataset used in this study is obtained from senior researchers who previously developed plant image–based medicinal plant identification systems. The dataset consists of leaf images of Indian medicinal plants captured under varying conditions. Each image is labeled with plant species information, which serves as genotype-linked reference labels, as plant species are associated with specific genetic compositions. These labels are later used to establish phenotype–genotype associations during model training and analysis.

2. Image Preprocessing

All collected leaf images are resized to a uniform resolution and normalized to maintain consistency across the dataset. Noise removal techniques are applied to improve image clarity. Background removal is performed to isolate the leaf region from the surrounding environment. Contrast enhancement using Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied to improve visibility of venation patterns. Additionally, vesselness filtering using the Meijering algorithm is employed to enhance vascular structures within the leaf.

3. Phenotypic Feature Learning

A multi-branch deep learning architecture is used to extract phenotypic features from the processed leaf images.

- A texture branch based on a CNN or Swin Transformer extracts color and texture-related features from RGB images.

- A venation branch learns topological features such as vein density, branching structure, and connectivity from enhanced grayscale images.

- A geometry branch extracts morphological shape descriptors such as leaf contour, curvature, aspect ratio, and symmetry from segmented leaf masks using a multilayer perceptron (MLP).The feature embeddings from all branches are concatenated and fused using a transformer encoder to obtain a shared phenotypic representation.

4. Phenotype Prediction

The fused phenotypic representation is passed through multiple classification and regression heads to predict phenotypic traits, including medicinal plant species classification and quantitative morphological characteristics. These predicted phenotypic traits represent observable plant characteristics derived from leaf morphology.

5. Genotype Association Using GWAS

Genome-Wide Association Study (GWAS) concepts are applied to associate predicted phenotypic traits with underlying genotype information. Known genotype data, including genes and single nucleotide polymorphisms (SNPs), are obtained from publicly available plant genome databases and existing biological literature. Statistical association analysis is performed to identify significant genotype–phenotype relationships.

6. Genotype Inference Model

The significant phenotypic features identified through GWAS are used as input to machine learning models such as Random Forest or XGBoost to infer genotype-associated markers. These models learn correlations between phenotypic patterns and genotype information, enabling prediction of candidate genes or SNPs responsible for observed medicinal plant traits.

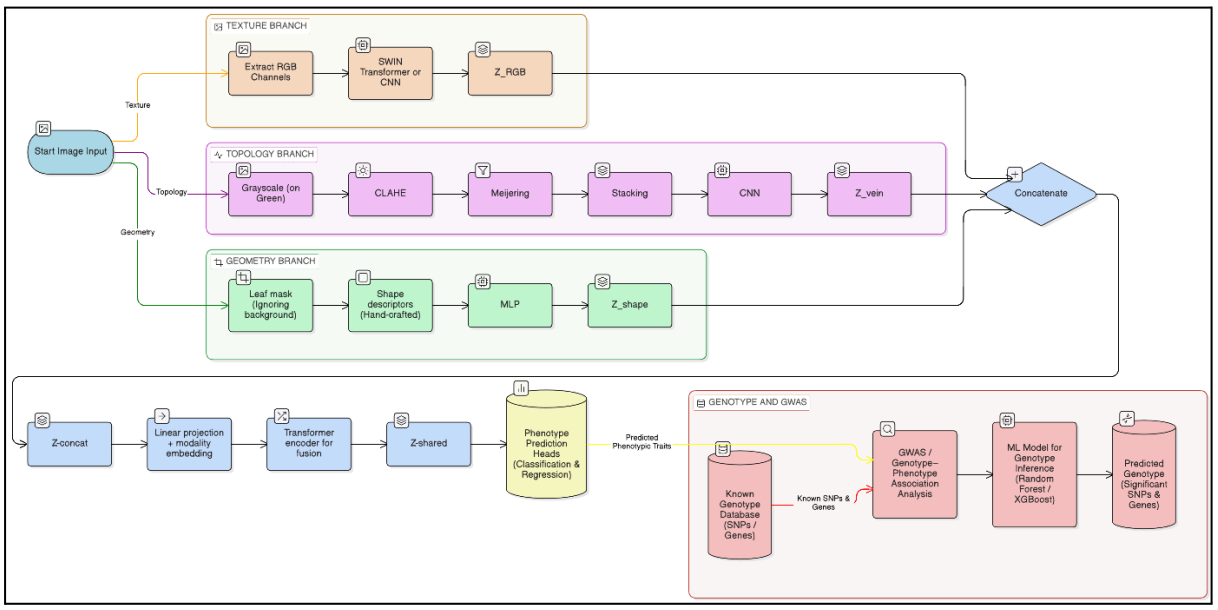
7. Model Training

The proposed deep learning architecture is trained using categorical cross-entropy loss for classification tasks and mean squared error loss for regression tasks. The Adam optimizer is used for optimization. Data augmentation techniques such as rotation, scaling, and flipping are applied to enhance model robustness and prevent overfitting.

8. Evaluation

The quality of extracted leaf venation patterns is evaluated using venation-specific metrics including edge density, vein coverage, skeleton length, branch points, and entropy. Edge density is used to assess noise and unwanted edge presence, while vein coverage measures the proportion of the leaf area identified as venation. Skeleton length and branch points evaluate the continuity and structural complexity of the venation network. Entropy is used to quantify the information content and balance between structured vein patterns and noise. The effectiveness of venation-enhanced phenotypic learning is further validated by analyzing its impact on phenotypic prediction and genotype inference, with results compared against known biological findings reported in the literature.

Block Diagram



Proposed Architecture

The preferred architecture is a transfer learning-based CNN such as ResNet-50 or EfficientNet, chosen for their ability to learn deep hierarchical features and handle complex venation patterns.

Architecture Flow

Leaf Image → Preprocessing → Venation Enhancement → CNN Feature Extraction → Genotype Mapping Layers → Predicted Genotype Class

Hardware, Software and Tools Requirements

Hardware

Component	Minimum Requirement	Recommended (For Heavy DL Models)	Purpose
Processor	Intel i5 / Ryzen 5	Intel i7 / Ryzen 7	Data preprocessing & pipeline control
RAM	8 GB	16–32 GB	Handling image batches & tensors
GPU	Optional	NVIDIA GPU (RTX series, CUDA enabled)	Training CNN/ViT models efficiently
Storage	50 GB SSD	100 GB SSD	Dataset + model checkpoints
System Type	64-bit OS	64-bit OS	DL framework compatibility

Software

Software / Tool	Role in System
Python	Core programming language
TensorFlow / PyTorch	Deep learning model development
OpenCV	Image processing & segmentation
NumPy	Numerical tensor operations
Pandas	Dataset handling
Matplotlib / Seaborn	Visualization & evaluation
Jupyter Notebook	Experimentation environment
CUDA & cuDNN	GPU acceleration
Git	Version control and collaboration

Proposed Evaluation Metrics

Venation Quality Evaluation Metrics :

1. Edge Density

The ratio of edge pixels to total pixels in the image, used to estimate the amount of noise and unwanted edges present in the venation extraction.

2. Vein Coverage

The proportion of image pixels classified as venation, indicating how much of the leaf area is detected as veins.

3. Skeleton Length

The total number of pixels in the skeletonized venation image, representing the continuity and extent of the vein network.

4. Branch Points

The number of junction points where veins split into multiple branches, reflecting the structural complexity of the venation pattern.

5. Entropy

A measure of the information content or randomness in the venation image, indicating the balance between structured vein information and noise.

Expected Outcomes

- Demonstration of genotype–phenotype correlation using leaf venation.
- Accurate inference of genotype-associated classes from phenotypic traits.
- Reduction in reliance on molecular genotyping techniques.
- A scalable, non-invasive framework for plant genetic characterization.

Conclusion

This project proposes a deep learning-based approach for inferring genotype-related information from leaf phenotypes using venation patterns. By leveraging CNN architectures and venation-driven feature learning, the system establishes a computational link between phenotype and genotype. While the approach does not replace molecular genotyping, it provides an efficient and scalable alternative for large-scale plant genetic analysis. The proposed framework has significant potential in medicinal plant authentication, biodiversity conservation, and plant breeding research.

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